Informatica

Informatica® PowerExchange for Hadoop
10.5.10

User Guide for PowerCenter

Informatica PowerExchange for Hadoop User Guide for PowerCenter
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Preface

Use the *Informatica® PowerExchange® for Hadoop User Guide for PowerCenter®* to learn how to read from or write to Hadoop. Learn to create a connection, develop mappings, and run sessions by using PowerCenter client.

Informatica Resources

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To find online support resources on the Informatica Network, visit https://network.informatica.com and select the eSupport option.

CHAPTER 1

Understanding PowerExchange for Hadoop

This chapter includes the following topics:

- PowerExchange for Hadoop Overview, 8
- Understanding Hadoop, 8
- PowerCenter and Hadoop Integration, 9

PowerExchange for Hadoop Overview

PowerExchange for Hadoop integrates PowerCenter with Hadoop to extract and load data.

You can connect a flat file source to Hadoop to extract data from Hadoop Distributed File System (HDFS). You can connect a flat file target to Hadoop to load data to HDFS. You can also load data to the Hive data warehouse system.

Understanding Hadoop

Hadoop provides a framework for distributed processing of large data sets across multiple computers. It depends on applications rather than hardware for high availability.

Hadoop applications use HDFS as the primary storage system. HDFS replicates data blocks and distributes them across nodes in a cluster.

Hive is a data warehouse system for Hadoop. You can use Hive to add structure to datasets stored in file systems that are compatible with Hadoop.

PowerCenter and Hadoop Integration

PowerExchange for Hadoop accesses Hadoop to extract data from HDFS or load data to HDFS or Hive.

To extract data from HDFS, a PowerExchange for Hadoop mapping contains a flat file source. To load data to HDFS or Hive, a PowerExchange for Hadoop mapping contains a flat file target.

In the Workflow Manager, you specify the HDFS flat file reader to extract data from HDFS. You specify the HDFS flat file writer to load data to HDFS or Hive. You select a Hadoop HDFS connection object to access the HDFS database tier of Hadoop.

The Integration Service communicates with Hadoop through the Java Native Interface (JNI). JNI is a programming framework that enables Java code running in a Java Virtual Machine (JVM) to call or be called.

CHAPTER 2

PowerExchange for Hadoop Configuration

This chapter includes the following topics:

- PowerExchange for Hadoop Configuration Overview, 10
- Configure PowerCenter for MapR , 10
- Registering the Plug-in, 13
- Update the Repository Plug-in, 13
- HDFS Connections for Hadoop Sources and Targets, 14
- Configure PowerCenter for Hadoop Cluster, 14

PowerExchange for Hadoop Configuration Overview

PowerExchange for Hadoop installs with PowerCenter.

Complete the following tasks to install or upgrade PowerExchange for Hadoop:

1. Install or upgrade PowerCenter.
2. If the Hadoop cluster runs MapR, configure PowerCenter to connect to a Hadoop cluster on MapR.
3. If you are upgrading from PowerCenter, recreate HDFS connections for updated Hadoop distribution versions.

Configure PowerCenter for MapR

You can enable PowerCenter to run mappings on a Hadoop cluster on MapR.

Perform the following tasks:

- Set environment variables for MapR.
- Copy MapR distribution files.
- Configure the PowerCenter Integration Service.

Environment Variables for MapR Distribution

When you use MapR distribution to access Hadoop sources and targets, you must configure environment variables.

For MapR Ticket Cluster

Configure the following MapR environment variables:

- Set environment variable MAPR_HOME to the following path: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>`.
- On the Linux operating system, change environment variable LD_LIBRARY_PATH to include the following path: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>`.
- Set the MapR Container Location Database name variable CLDB in the following file: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>/conf/mapr-clusters.conf`.
- Optional. Set environment variable MAPR_TICKETFILE_LOCATION to the location of the MapR ticket file.

For MapR Kerberos Cluster

Configure the following MapR environment variables:

- Set environment variable MAPR_HOME to the following path: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>`.
- On the Linux operating system, change environment variable LD_LIBRARY_PATH to include the following path: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>`.
- Set the MapR Container Location Database name variable CLDB in the following file: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>/conf/mapr-clusters.conf`.
- Set environment variable JAVA_OPTS to: `-Dhadoop.login=hybrid -Dhttps.protocols=TLSv1.2`
- Optional. Set environment variable MAPR_TICKETFILE_LOCATION to the location of the MapR ticket file.
- Optional. Set environment variable KRB5_CONFIG to the following path: `<Informatica installation directory_DMAPR>/java/jre/lib/security/krb5.conf`

For MapR non-Secure Cluster

Configure the following MapR environment variables:

- Set environment variable MAPR_HOME to the following path: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>`.
- On the Linux operating system, change environment variable LD_LIBRARY_PATH to include the following path: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>`.
- Set the MapR Container Location Database name variable CLDB in the following file: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>/conf/mapr-clusters.conf`.

Copy MapR Distribution Files for PowerCenter Mappings in the Native Environment

When you use the MapR distribution to run mappings in a native environment, you must copy MapR files to the machine on which the PowerCenter Integration Service runs.

Perform the following steps:

1. Create a directory where PowerCenter administrator user has the read access. For example: `<infa_home>/pwx-hadoop/conf`
2. Go to the following directory on any node in the cluster: `<MapR installation directory>/conf`
   For example, go to the following directory: `/opt/mapr/conf`.
3. In the Administrator tool, go to the **Services and Nodes** tab. Select the **Processes** view for the required PowerCenter Integration Service and add the environment variable "CLASSPATH" with the value of the directory created in step 1.
4. In the Domain Navigator, select the PowerCenter Integration Service.
5. Recycle the Service.
   Click **Actions** > **Recycle Service**.

Configure the PowerCenter Integration Service

To enable support for MapR, configure the PowerCenter Integration Service.

Perform the following steps:

1. Log in to the Administrator tool.
2. Click the **Services and Nodes** tab.
3. In the Domain Navigator, select the PowerCenter Integration Service.
4. Click the **Processes** view.
5. Add the environment variables as required. For more information about environment variables, see "Environment Variables for MapR Distribution" on page 11.
6. Recycle the Service.

   Click **Actions** > **Recycle Service**.

Install the MapR Client and Configure the Environment Path

To run mappings on a MapR 5.2 with MEP 3.x cluster, you must install the MapR client and then configure environment settings on the Informatica domain machine.

Perform this task in the following situations:
- You are integrating for the first time. - You upgraded from version 10.2 or earlier.

Property	Description
MAPR_HOME	**Note:** Configure the MAPR_HOME property in the Environment Variables section of the PowerCenter Repository Service.
Java JDK ClassPath	Set the Java SDK classpath. The PowerCenter Integration Service appends the values that you set to the system CLASSPATH. For example: `<Informatica installation directory>/source/services/shared/hadoop/mapr<version>/lib/*:/opt/mapr/lib/*`

Registering the Plug-in

A plug-in is an XML file that defines the functionality of PowerExchange for Hadoop. To register the plug-in, the repository must be running in exclusive mode. Use Informatica Administrator (the Administrator tool) or the pmrep RegisterPlugin command to register the plug-in.

Register the plug-in if you are upgrading from a previous PowerCenter release.

The plug-in file for PowerExchange for Hadoop is pmhdfs.xml. When you install the Repository component, the installer copies pmhdfs.xml to the following directory:

```
<Informatica installation directory>/server/bin/native
```

Note: If you do not have the correct privileges to register the plug-in, contact the user who manages the PowerCenter Repository Service.

Update the Repository Plug-in

If you upgraded an existing repository, you must update the repository plug-in to enable PowerExchange for HDFS to run on the Hadoop distribution. If you created a new repository, skip this task.

1. Ensure that the Repository service is running in exclusive mode.
2. On the server machine, open the command console.
3. Run cd <Informatica installation directory>/server/bin
4. Run ./pmrep connect -r <repo_name> -d <domain_name> -n <username> -x <password>
5. Run ./pmrep registerplugin -i native/pmhdfs.xml -e -N true
6. Set the Repository service to normal mode.

7. Open the PowerCenter Workflow manager on the client machine.

   The distribution appears in the Connection Object menu.

HDFS Connections for Hadoop Sources and Targets

Use a HDFS connection object to access a Hadoop source or target.

When you upgrade PowerExchange for Hadoop, you must recreate HDFS connections to access Hadoop source or target. Use the Namenode URI property to recreate the HDFS connections.

Configure PowerCenter for Hadoop Cluster

You can configure PowerCenter and the PowerCenter Integration Service to read data from and write data to a Hadoop cluster. The Hadoop cluster could be a High Availability (HA), non-HA, kerberos-enabled, or non-kerberos cluster.

Perform the following steps to configure PowerCenter for Cloudera, Hortonworks, IBM BigInsights, and MapR distributions:

1. On the Informatica node where PowerCenter Integration Service runs, create a directory. The PowerCenter administrator user must have the read access on this directory. For example: `<INFA_HOME>/pwx-hadoop/conf`
2. Copy the following files from Hadoop cluster to directory created in step 1:
   - `/etc/hadoop/conf/core-site.xml`
   - `/etc/hadoop/conf/mapred-site.xml`
   - `/etc/hadoop/conf/hdfs-site.xml`
   - `/etc/hive/conf/hive-site.xml`
3. Optional. Applicable to kerberos-enabled clusters. Ensure that the PowerCenter administrator user exists on all Hadoop cluster nodes and has the same UID and run kinit to create Kerberos ticket cache file on all nodes.
4. Optional. Applicable to kerberos-enabled clusters. Run the kinit on the Informatica node where PowerCenter Integration Service runs to create the Kerberos ticket cache file. For example: `/tmp/krb5cc_<UID>`
5. Optional. Applicable to kerberos-enabled clusters except MapR. Edit the `core-site.xml` configuration set in the directory created in step 1 and add the following parameter:

   ```
   <property>
   <name>hadoop.security.kerberos.ticket.cache.path</name>
   <value>/tmp/REPLACE_WTH_CACHE_FILENAME</value>
   <description>Path to the Kerberos ticket cache. </description>
   </property>
   ```
6. In the Administrator tool, go to the **Services and Nodes** tab. Select the **Processes** view for the required PowerCenter Integration Service and add the environment variable "CLASSPATH" with the value of the directory created in step 1.

7. Recycle the Service.

   Click **Actions** > **Recycle Service**.

8. In the Workflow Manager, create the HDFS connection and assign to source or target and run the workflow. When you create the HDFS connection, use the value for the `fs.default.name` property for the NameNode URI. You can find the value for the `fs.default.name` property in the `core-site.xml` file.

CHAPTER 3

PowerExchange for Hadoop Sources and Targets

This chapter includes the following topic:

- PowerExchange for Hadoop Sources and Targets Overview, 16

PowerExchange for Hadoop Sources and Targets Overview

You include a flat file source definition in a mapping to extract Hadoop data. You include a delimited flat file target definition in a mapping to load data into HDFS or to a Hive table.

You can import a flat file definition or manually create one. To load data to a Hadoop target, the flat file definition must be delimited.

CHAPTER 4

PowerExchange for Hadoop Sessions

This chapter includes the following topics:

- PowerExchange for Hadoop Sessions Overview, 17
- PowerExchange for Hadoop Connections, 17
- Sessions with Hadoop Sources, 18
- Sessions with Hadoop Targets, 21

PowerExchange for Hadoop Sessions Overview

After you create a PowerExchange for Hadoop mapping in the Designer, you create a PowerExchange for Hadoop session in the Workflow Manager to read, transform, and write Hadoop data.

Before you create a session, configure a Hadoop HDFS application connection to connect to the HDFS host. When the Integration Service extracts or loads Hadoop data, it connects to a Hadoop cluster through the HDFS host that runs the name node service for a Hadoop cluster.

If the mapping contains a flat file source, you can configure the session to extract data from HDFS. If the mapping contains a flat file target, you can configure the session to load data to HDFS or a Hive table.

When the Integration Service loads data to a Hive table, it first loads data to HDFS. The Integration Service then generates an SQL statement to create the Hive table and load the data from HDFS to the table.

PowerExchange for Hadoop Connections

Use a Hadoop HDFS application connection object for each Hadoop source or target that you want to access.

You connect to a Hadoop cluster through an HDFS host that runs the name node service for a Hadoop cluster.

The following table describes the properties that you configure for a Hadoop HDFS application connection:

Property	Description
Name	The connection name used by the Workflow Manager. Connection name cannot contain spaces or other special characters, except for the underscore character.
User Name	The name of the user in the Hadoop group that is used to access the HDFS host.
Password	Password to access the HDFS host. Reserved for future use.

HDFS Connection URI	The URI to access HDFS. Use the value for the `fs.default.name` property for the NameNode URI. You can find the value for the property for the NameNode URI. You can find the value for the `fs.default.name` property in the `core-site.xml` configuration set.
	Syntax for Hadoop distributions:
	`hdfs://<namenode>:<port>`
	Where
	- <namenode> is the host name or IP address of the NameNode.
	- <port> is the port that the NameNode listens for remote procedure calls (RPC).
	Syntax for the MapR distribution:
	`maprfs:///`
	Syntax for the HDInsight distribution:
	- `adl:// <nameservices>`
	- `wasb://<nameservices>`

Hive Driver Name	The name of the Hive driver. By default, the driver name is: `org.apache.hive.jdbc.HiveDriver`

Hive URL	The URL to the Hive host.
	For MapR Ticket cluster, specify the URL in the following format:
	`jdbc:hive2://hostname:portnumber/default;auth=MAPRSASL/default;auth=MAPRSASL`
	For MapR Kerberos cluster, specify the URL in the following format:
	`jdbc:hive2://hostname:portnumber/default;auth=MAPRSASL/default;principal=<spn>`

Hive User Name	The Hive user name. Reserved for future use.
Hive Password	The password for the Hive user. Reserved for future use.
Hadoop Distribution	The name of the Hadoop distribution. Default is cloudera_cdh.

Sessions with Hadoop Sources

You can configure a session to extract data from HDFS.

When you configure a session for a Hadoop source, you select the HDFS Flat File reader file type and a Hadoop HDFS application connection object. You can stage the source data and configure partitioning.

Staging HDFS Source Data

You can optionally stage an HDFS source. The Integration Service stages source files on the local machine and then loads data from the staged file or files into the target.

Stage an HDFS source when you want the Integration Service to read the source files and then close the connection before continuing to process the data.

Configure staging for an HDFS source by setting HDFS flat file reader properties in a session.

You can configure the following types of staging:

Direct

Use direct staging when you want to read data from a source file. The Integration Service reads data from the source file and stages the data on the local machine before passing to downstream transformations.

For example, if you stage a file named `source.csv` from the Hadoop source location to the following directory:

```
c:\staged_files\source_stage.csv
```

The Integration Service stages `source.csv` as `source_stage.csv` in the `c:\staged_files` directory. Then, the Integration Service loads data from the `source_stage.csv` file into the Hadoop target file as specified by the output file path in the HDFS flat file writer properties.

Indirect

Use indirect staging when you want to read data from multiple source files. The Integration Service reads data from multiple files in the source. It creates an indirect file that contains the names of the source files. It then stages the indirect file and the files read from the source in the local staging directory before passing to downstream transformations.

For example, you stage the files named `source1.csv` and `source2.csv` from the Hadoop source location to the following directory:

```
c:\staged_files\source_stage_list.txt
```

The Integration Service creates an indirect file named `source_stage_list.txt` that contains the following entries:

```
source1.csv
source2.csv
```

The Integration Service stages the indirect file and the source files. In the `c:\staged_files` directory, you would see the following files:

```
source_stage_list.txt
source1.csv
source2.csv
```

Then, the Integration Service loads data from the staged source files into the Hadoop target path as specified by the output file path in the HDFS flat file writer properties.

To configure staging, set the following session properties:

Property	Set to
Is Staged	Enabled.
File Path	For direct staging, enter the name and path of the source file. For indirect staging, enter the directory path to the source files.
Staged File Name	Name and path on the local machine used to stage the source data.

Hadoop Source Connections

To extract data from a Hadoop source, you configure the session properties to select the HDFS flat file reader file type and a Hadoop HDFS application connection object.

In the sources node section of the **Mapping** tab in the session properties, select **HDFS Flat File Reader** as the reader type for the source. Then, select a Hadoop HDFS application connection object for the source.

When you select HDFS flat file reader type, you can select a code page of a delimited file from the codepage drop-down list. You cannot set the code page to use a user-defined variable or a session parameter file.

Session Properties for a Hadoop Source

You can configure a session for a Hadoop source to set staging and partitioning properties.

The following table describes the properties that you can configure for a Hadoop source:

Session Property	Description
Is Staged	Before reading the source file, the Integration Service stages the remote file or files locally.
Staged File Name	The local file name directory where the Integration Service stages files. If you use direct staging, the Integration Service stages the source file using this file name. If you use indirect staging, the Integration Service uses the file name to create the indirect file.
Concurrent read partitioning	Order in which multiple partitions read input rows from a source file. You can choose one of the following options: - **Optimize throughput**. The Integration Service does not preserve row order when multiple partitions read from a single file source. Use this option if the order in which multiple partitions read from a file source is not important. - **Keep relative input row order**. The Integration Service preserves the input row order for the rows read by each partition. Use this option to preserve the sort order of the input rows read by each partition. - **Keep absolute input row order**. The Integration Service preserves the input row order for all rows read by all partitions. Use this option to preserve the sort order of the input rows each time the session runs. In a pass-through mapping with passive transformations, the order of the rows written to the target will be in the same order as the input rows.
File Path	The Hadoop directory path of the flat file source. The path can be relative or absolute. If relative, it is relative to the home directory of the Hadoop user. For example, you might specify the following value in the **File Path** property: `\home\foo\`

Sessions with Hadoop Targets

You can configure a session to load data to HDFS or a Hive table.

When you configure a session for a Hadoop target, you select the HDFS Flat File writer file type and a Hadoop HDFS application connection object.

When you configure the session to load data to HDFS, you can configure partitioning, file header, and output options. When you configure the session to load data to a Hive table, you can configure partitioning and output options.

When the Integration Service loads data to a Hive table, it generates a relational table in the Hive database. You can overwite the Hive table data when you run the session again.

Hadoop Target Connections

To load data to a Hadoop target, you configure the session properties to select the HDFS flat file writer file type and a Hadoop HDFS application connection object.

In the targets node section of the **Mapping** tab in the session properties, select **HDFS Flat File Writer** as the writer type for the target. Then, select a Hadoop HDFS application connection object for the target.

When you select HDFS flat file writer type, you can select a code page of a delimited file from the codepage drop-down list. You cannot set the code page to use a user-defined variable or a session parameter file.

Hadoop Target Partitioning

When you configure a session to load data to a Hadoop target, you can write the target output to a separate file for each partition or to a merge file that contains the target output for all partitions.

You can select the following merge types for Hadoop target partitioning:

No Merge

The Integration Service generates one target file for each partition. If you stage the files, the Integration Service transfers the target files to the remote location at the end of the session. If you do not stage the files, the Integration Service generates the target files at the remote location.

Sequential Merge

The Integration Service creates one output file for each partition. At the end of the session, the Integration Service merges the individual output files into a merge file, deletes the individual output files, and transfers the merge file to the remote location.

If you set the merge type to sequential, you need to define the merge file path and the output file path in the session properties. The merge file path determines the final Hadoop target location where the Integration Service creates the merge file. The Integration Service creates the merge file from the intermediate merge file output in the location defined for the output file path.

Session Properties for a Hadoop Target

You can configure a session for a Hadoop target to load data to HDFS or a Hive table. When you load data to a Hadoop target, you can set partitioning properties and file paths. When you load data to HDFS, you can also set header options.

The following table describes the properties that you can configure for a Hadoop target:

Session Property	Description
Merge Type	Type of merge that the Integration Service performs on the data for partitioned targets. You can choose one of the following merge types: - No Merge - Sequential Merge
Append if Exists	Appends data to a file. If the merge file path refers to a directory, this option applies to the files in the directory.
Header Options	Creates a header row in the flat file when loading data to HDFS.
Auto generate partition file names	Generates partition file names.
Merge File Path	If you choose a sequential merge type, defines the final Hadoop target location where the Integration Service creates the merge file. The Integration Service creates the merge file from the intermediate merge file output in the location defined in the output file path.
Generate And Load Hive Table	Generates a relational table in the Hive database. The Integration Service loads data into the Hive table from the HDFS flat file target.
Overwrite Hive Table	Overwrites the data in the Hive table.
Hive Table Name	Hive table name. Default is the target instance name.
Externally Managed Hive Table	Loads Hive table data to the location defined in the output file path.
Output File Path	Defines the absolute or relative directory path or file path on the HDFS host where the Integration Service writes the HDFS data. A relative path is relative to the home directory of the Hadoop user. If you choose a sequential merge type, defines where the Integration Service writes intermediate output before it writes to the final Hadoop target location as defined by the merge file path. If you choose to generate partition file names, this path can be a directory path.
Reject File Path	The path to the reject file. By default, the Integration Service writes all reject files to service process variable directory, $PMBadFileDir.

CHAPTER 5

Data Type Reference

This chapter includes the following topics:

- Data Type Reference Overview, 23
- Flat File and Transformation Data Types, 23

Data Type Reference Overview

PowerCenter uses the following data types for Hadoop data objects:

- Flat file data types. Flat file data types appear in the physical data object column properties.
- Transformation data types. Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the PowerCenter Integration Service uses to move data across platforms. Transformation data types appear in all transformations in a mapping.

When the PowerCenter Integration Service reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When the PowerCenter Integration Service writes to a target, it converts the transformation data types to the comparable native data types.

Flat File and Transformation Data Types

Flat file data types map to transformation data types that the PowerCenter Integration Service uses to move data across platforms.

The following table compares flat file data types to transformation data types:

Flat File Data type	Transformation Data type	Range
Bigint	Bigint	Precision of 19 digits, scale of 0
Datetime	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. (precision to the nanosecond)
Double	Double	Precision of 15 digits
Int	Integer	-2,147,483,648 to 2,147,483,647

Flat File Data type	Transformation Data type	Range
Nstring	String	1 to 104,857,600 characters
Number	Decimal	Precision 1 to 28, scale 0 to 28 If the precision is greater than 15, the Data Integration Service converts decimal values to double in low-precision mode.
String	String	1 to 104,857,600 characters
TimestampWithTZ	TimestampWithTZ	Aug. 1, 1947 A.D to Dec. 31, 2040 A.D. -12:00 to +14:00 Precision of 36 and scale of 9. (precision to the nanosecond) Timestamp with Time Zone data type does not support the following time zone regions: - AFRICA_CAIRO - AFRICA_MONROVIA - EGYPT - AMERICA_MONTREAL **Note:** TimestampWithTZ is applicable only to the native environment.

When the PowerCenter Integration Service reads non-numeric data in a numeric column from a flat file, it drops the row and writes a message in the log. Also, when the PowerCenter Integration Service reads non-datetime data in a datetime column from a flat file, it drops the row and writes a message in the log.

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